IIT Madras

ONLINE DEGREE

Database Management Systems
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Why DBMS? /1

Welcome back to Module 2 of database management systems course in the IIT Madras online BSc program. In Module 1, we have taken a very basic look at how databases have pervaded all aspects of our life, touch any part, touch any application or even when you are not aware, I mean just to tell you the extreme in a way, even for people who just use a basic cell phone to make calls, you are using an extensively large database application which not only keeps track of the numbers, the location tracks, the connection track and so on of the caller and the callee and the overall infrastructure of towers and base stations and all of that.

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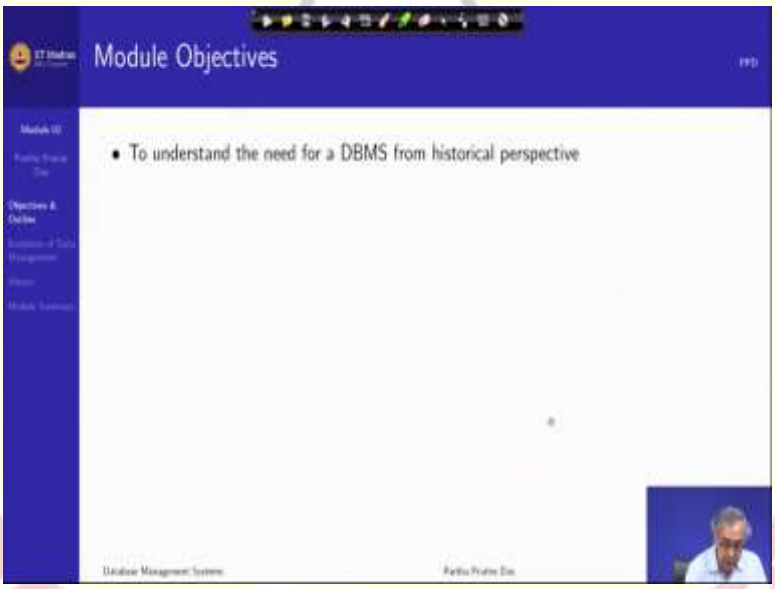

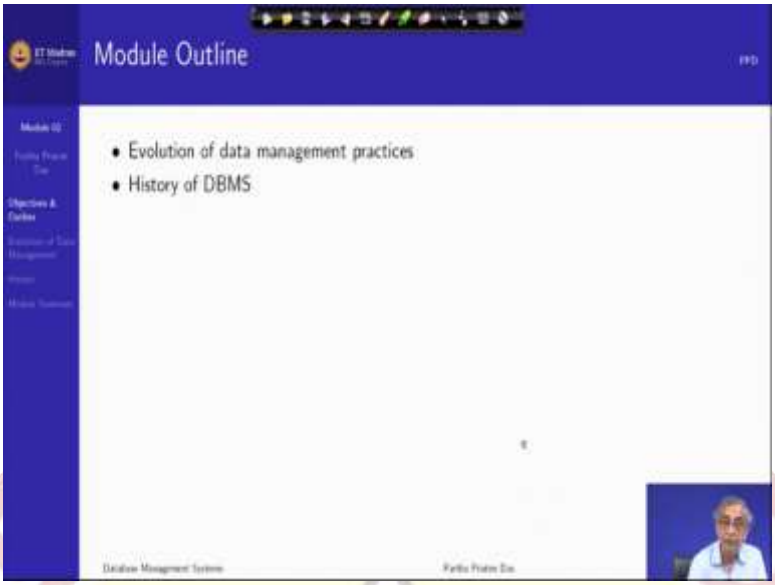

So, we cannot live without databases today. In the current module, we again in the current module and the next, which we have titled as part 1 and 2 of why DBMS, we take a little deeper look in two perspectives. In this module, what we want to do is to understand the need for a database management system for somewhat of a historical perspective. Asking ourselves fundamentally what is that database management systems are needed for?

What did we do say 300 years back 400 years back, what did we do 100 years back, what did we do 30 years back, what did we do 10 years back and why and how the database management systems are evolving, keeping two sides in balance, one is, one side is the availability of technology, we are talking about a time where technology is the key thing which keeps on moving makes more things available, and hand in hand with that on the other side are the way the society is transforming.

We have been part of a big pandemic. And you know, we all know in one of the ways or in some of the ways as a nation we have tried to respond to it was to use say an application called Aarogya Setu. Aarogya Setu is an application to find out about your health condition related to Coronavirus and the health conditions of people potentially near to you.

So, you can think of that these are people who do not even know. But it is possible to track and correlate this information. So, what kind of huge database go behind that, which is keeping track of the conditions of several people who use the Aarogya Setu App, and their locations and their movements and all.

The other that all of you are aware of in this context is the COWIN App which has been tracking the entire nation's vaccination program. It is tracking who is getting vaccinated,

where is she or he getting vaccinated? What vaccine has been administered, what is the first dose or the second dose, a date when is the next vaccine to be given the other card or passport of that person and finally the certification. Another huge life changing database application.

Now, even if you look back 10 years ago, this were not possible. We did not have Aadhaar cards we did not have this kind of network. We did not have this kind of scalability of database systems. So, in this module, we try to take a little historical look on how have we evolved as a civilisation. In terms of data, and records management, and in the next we will talk about specifically about why you need separate kind of systems for solving data related problems, why does not simple computer programs can solve this problem. So, we will talk about evolution of data management practices and a little bit of history of the DBMS.

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So, first of all the evolution of data management. So, if we ask in principle, then what does the management of data or which earlier was used to be called management of records? What all are needed in the human society? What is the basic need? The first of course is storage, you need to remember things, how much money I have, how many inventory I control, how much of how many people an organization employs, and so on. So, storage is important.

Along with the storage comes retrieval, that I should not be whatever data is stored, I should be able to get it back. So, storage and retrieval are the key requirements of data and records management. I am not specifically talking about database systems, as you know today, but I am saying the need of management of data and management of records of different things.

Then you have transactions, transactions is where something is being exchanged, something is being done on that data, either certain part moving from one place to the other or one, so called account to another like I go to a showroom and buy a two wheeler. So, a transaction has happened, a two-wheeler whose ownership was with the dealer, or was with the company through the dealer is now transferred to my name. So, this is a transaction. And accordingly corresponding to that there are financial transactions, the money that I had to pay to the company through the dealer. So, transactions are a very important part.

The fourth is audit, you always need to know what happened, did things happen properly? Was there any mistake? Are all the rules and regulations of that particular segment properly followed? Was tax paid? On the sale of the two-wheeler? Was, why are all these financial scams we are looking at? Because it is possible to dig out from the stored data with the appropriate retrieval and audit to see what who did certain things may not be right. You have archival purposes, that you need to just keep that data for maybe some possible future use. There could be more, but I just talked about some of the salient requirements.

So, we will have to the reason I am putting I know you, all of you know this, all of you know this I am not saying anything new. But I just want to highlight these points because you have to always keep these in perspective, when you learn this course. This is your, these are your challenge points. You are trying to make a system, you are trying to improve a system you are trying to make changes to a system keeping these kinds of requirements in mind.

And obviously, related question is whom do you do it for? You could do it for an individual, for me for you, for your friend for your parents, or it could be for a small enterprise, small organization or for a big enterprise like Google, like Reliance Industries, like Tata Motors, so

on. Or it could be for a global organization, it could be for United Nations, it could be for any other country's government or in more global organizations like Apple.

So, what do you have to do? What we need and for whom we need are very clear. The question is what should be the approach, and if you look at the human history, there has been primarily two major approaches in this practice, which are very natural. One is physical. So, the world started physical. So, from the day people started realizing the need for maintaining records to satisfy storage, retrieval, transaction, audit, archival these kinds of requirements. People started maintaining records, physically. And then came the electronic era.

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So, most of the time, the human civilization has been around, we have seen physical forms of data and records management, which is in certain parlance, very formally known as bookkeeping. If we if any of you have familiarity with accounting courses and so on, you will know that there are courses like accountancy and bookkeeping, which basically talks about how you keep the physical Ledger's, how are you… I mean what in what style you put the data in it, what style you write journals, that is notes and so on.

Now, if you go back couple of centuries, you will find that well, this practice is coming from time immemorial. The practice of having ledgers and journals, but things started streamlining somewhat towards the end of nineteenth century when an American inventor by the name of Henry Brown, first invented something which is called respectable… receptacle for storing and preserving papers, which in today's time, we know as file cabinet, how to organize and have had a severe evolution based on that.

There is a significant correspondence between this physical organization of the data to the electronic mimicry of that in terms of the database system. So, that transfer knowledge is very important to realize, even in terms of memory, instant terms of, writing down everything is laborious as well as error prone. So, Hollerith, Herman Hollerith, who kind of started using adapting punch cards that were used in the Jacquard loom for weaving as memory so that you could do very simple mechanical computation based on these ledgers based on this ledger data, that also happened somewhere around towards the end of nineteenth century.

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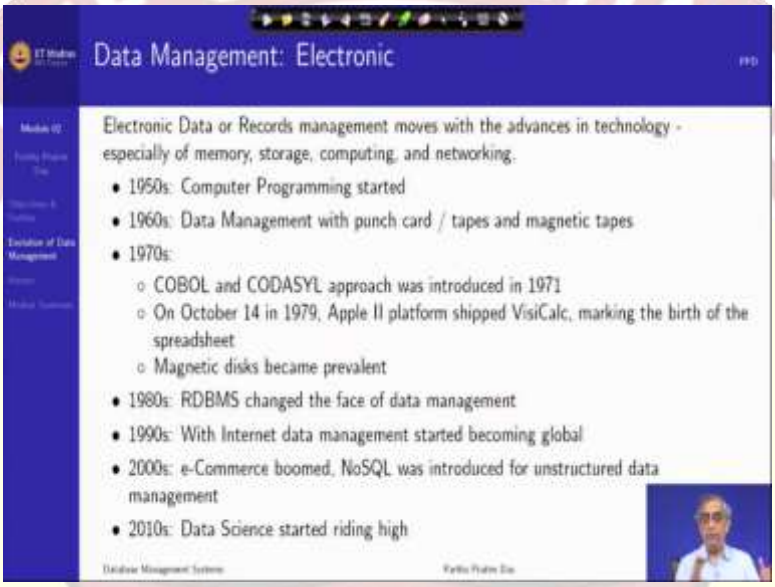

And then, so on. I mean before that, several centuries, people followed the same practice following the getting into the twentieth century also the same practices continued, till you came to almost the middle of the twentieth century when computer programming started, computer started happening, programming started and in 60s mostly, you started seeing the database management system or electronic management of data, where you need storage, retrieval, transaction, audit, archival this kind of.

So, for storage, which is the primary thing because you have to keep the records where punch cards were used long punch tapes, I do not know if any one of you have seen, or were used then it evolved to magnetic tapes and so on, went on for about 10-15 years till in 70s certain significant improvements and development happens in terms of COBOL language being introduced, it is still used, 50 years on it is still used.

CODASYL approach that is one approach of doing data management came in, Apple first introduced VisiCalc, VisiCalc is kind of the forefather or great grand forefather of our

spreadsheets today, Excel or Google Sheet as easy, magnetic disks became prevalent and so on. Then in 80s, we had relational database systems which changed the face of data management in 90s. Internet came in so that started making the whole data and records management global.

In 2000s, ecommerce really started booming, all those most of the significant e commerce activities you see today, they started around 2000s and this some died, some flourished. And as that kept on happening, there was more and more need and more and more requirement to manage unstructured data, data which is not just textual data, which is just not numbers, data about images, the Facebook profile photos, data about videos, data about audio tracks and so on so forth. Data about just freely written natural language, text and so on. And no SQL was born.

In the last decade data science started riding high, and today you are there you will write the next future of data management using electronic systems. So, let us this is a very broad evolution if you if you look at it.

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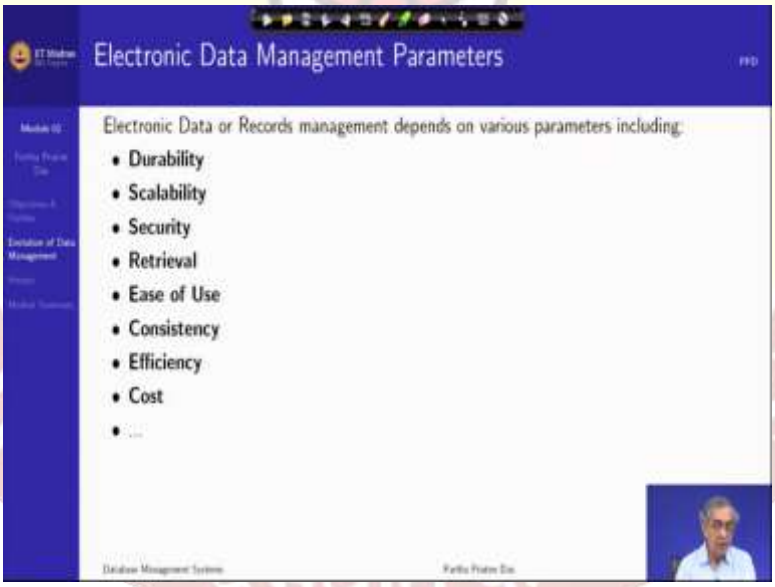

Now, in this whole process, there are several parameters on which this data and records management system particularly in the electronic form has to take care, things like durability and store a data it must be available for ages together that was a basic, drawback of the physical system, papers were out the termite will eat it up, or even the basic acid in the paper will cause the decay of the paper on which you kept the records but electronic records must be more durable. Are they? Is a big question.

And that will have to be ensure, how do you ensure that they remain durable as well. It must be scalable; it should work for say 10-20-100 people as well as 10,000; 100,000 millions billions of people or transactions and so on. It has to be absolutely secured. Of course, depending on the application, all of these will have a certain operating range, the security of say the different information systems that are associated with a difference.

And the security of a digital library are not equivalent. A digital library needs some security so that copyright is stuff and not stolen. But that security will be far less compared to the security of where the ward management is done. So, though but those are those are aspects these aspects we will have to always keep in mind keep questioning about them, what is the retrieval? How quickly you need to retrieve, is it okay to get your data within a day within a couple of seconds under millisecond, under nanosecond and so on so forth.

It should be easy to use of course, otherwise, it has not served the basic purpose consistency, we talked a little bit about consistency in the last class that always if I do something on that database, if I do some transaction, then before the transaction and after the transaction, the database must be in a consistent state it should not be that of, a transaction has been done to make a payment of 100 rupees to my friend and in that process 100 rupees get disappears from the system. That is not certainly acceptable.

Efficiency has to be high that efficiency as a whole not only of the is not only the time efficiency, it must be that is what the speed of retrieval is, but efficiency in terms of you know, amount of storage requirement maintenance requirement, efficiency and productivity of the staff those are involved, the programmers those are involved because all of these lead to the cost considerations. So, these are some of the not this is not a complete list the some of the key parameters on which data and record management's need to be looked at.

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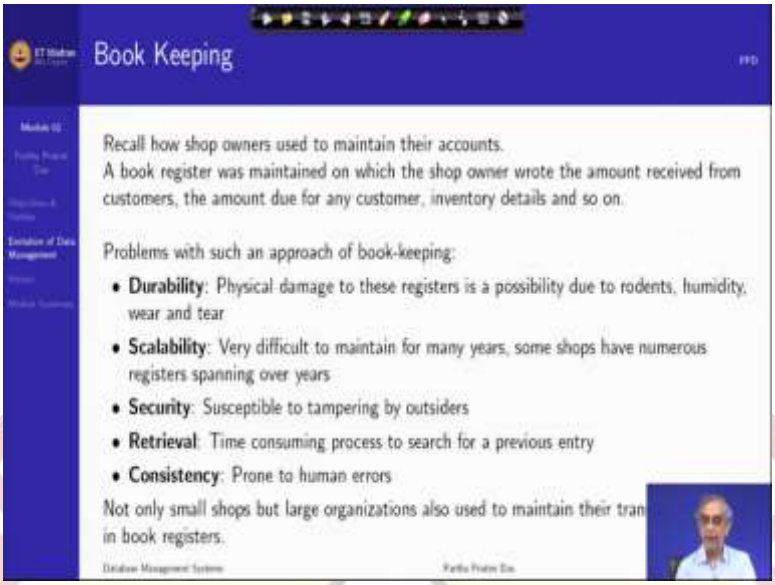

So, if we just you know, more a fun discussion, if we just look back into bookkeeping, you know, physical records management, widely used everywhere till today, you go to a grocery shop, many a grocery shop have a small PC, some have migrated, and they are more savvy, they are migrated to doing business on mobile phone, but there are several who will just write it down.

This is you go to particularly you go to a medicines shop, they give you the medicine and then they see that this particular they are low on that stock, they will take out a ledger right down on the journal that this medicine codopyrine is not there in stock and so on. So, if we recall on that, then we will know we have seen that. What are the problems? There are several problems one durability is a problem physical damage. I mean it can very easily get damaged can get as we say with rodent, termite, humidity, wear and tear, fire there are a lot of ways physical records might get destroyed. Scalability is very difficult to maintain.

You can manage a medicine store for maybe 100 customers, 200 customers on ledger you cannot maintain a medicine store of a million customers on paper. It is not possible. Security. Always because you always have to have trusted people you need physical lock and key and so on so forth, you know, big saves, and all that for maintaining the physical records.

Retrieval is obviously time consuming; you will have to look up, take out that big stuff from the file cabinet look up and so on. Consistency is certainly you can, it is quite possible that you have made a payment that has been received but has not been entered on your part of the ledger. So, there are several problems of this physical bookkeeping systems.

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Somewhat of a better solution, which many of us is widely used, and kind of it came in about 40 years back. I just mentioned VisiCalc. So, that was about 40 years back. So, it is a spreadsheet software which allow you to keep records in simple terms, you have paid applications like Excel or you can use Google Sheet which is free. And it these also have different issues, these are, but these are less prone to durability issues of damage, because you can keep a Google Sheet in a Google Cloud which has a lot of backend infrastructure to make sure that it does not get lost.

Then there is scalability is easier to search, you can password protected is not a very good protection, there are several, you know systems available by which it can be broken, but well, some sort of protection can be there, it is easy to use, consistency of course, is not much guaranteed in spreadsheets, you will have to either write lots of formula or kind of make sure that you are using it in the right way.

But it is better than a physical ledger and journal-based system, which according to these criteria, can access it based on more criteria, you can do it yourself. So, this is still used, I mean this is widely used not still this is widely used for particularly single users or for small enterprise applications. A good example of how a moderate range data management solution can provide.

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But certainly, these are all spreadsheets or CVS file CSV files and so on are all, basically file system based, we say these are flat file systems, flat file solutions. So, they have lot of limitations as well. If you they are scalability is limited, like spreadsheets have a maximum limit on the number of rows. Consistency management, we said is relatively a big challenge.

If we have certain constraints on the data, it is not easy I mean you can you can try in Excel then you will see that Excel does allow you to put some constraints, but they are just data entry constraints that is, you cannot enter a negative value in this cell. But if you have constraints resulting from transactions.

It is very difficult to manage that in a file-based system because you will have to write separate code for doing each one of that, it permission levels are limited, you can give a password but you cannot have multiple people use the same file in multiple levels of authority. So, these are some of the typical limitations in the evolution which finally makes sure that the only solution is a database management system which we are going to study here.

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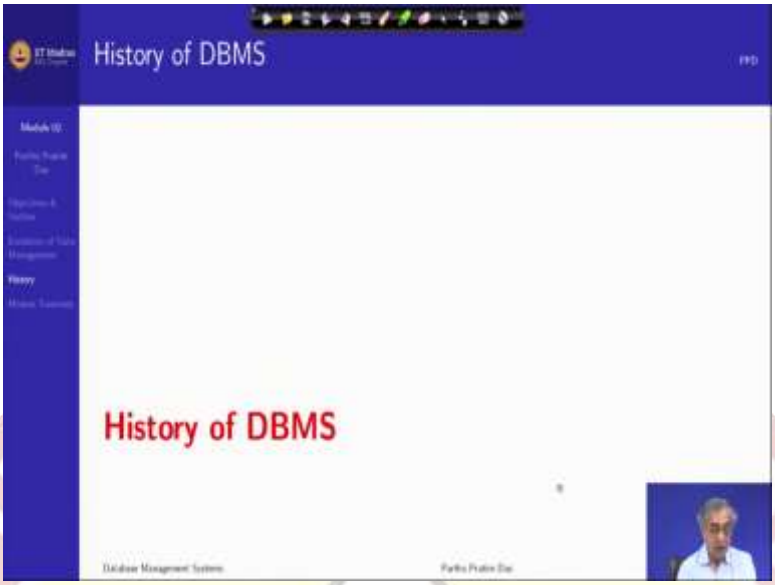

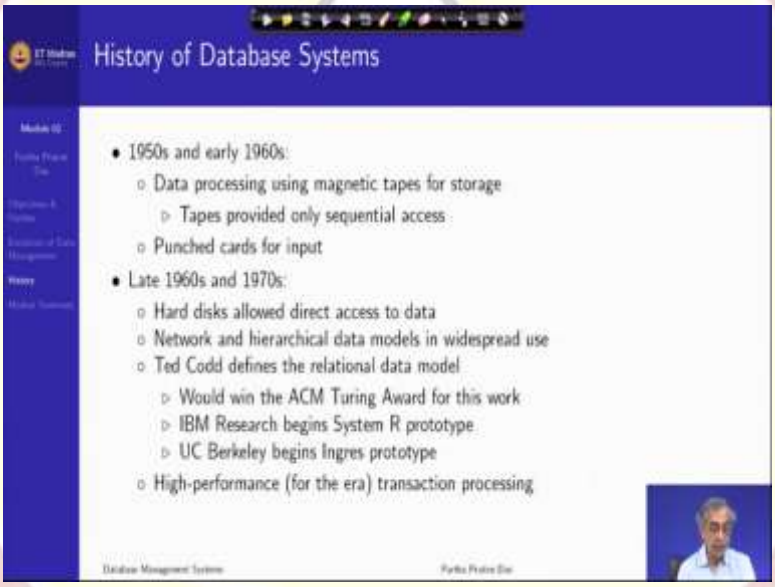

Talking a little about the history I will be very brief here because it is a long and deep history. As I said it started between 1950s and 60s, tapes and punch cards, then in late 1960s and 70s hard disks magnetic disks came in and these are some of the people whose work you are going to read about like Ted Codd, who define the relational data model we will learn about Boyce Codd normal form and so on.

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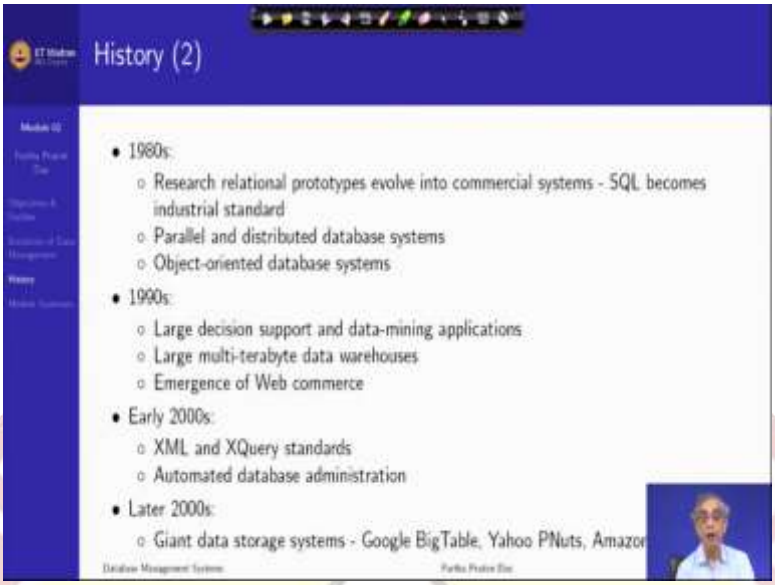

And slowly the, evolution towards more and more sophisticated data management systems started happening. 80's really saw a boom, through which SQL which we will learn about start learning about very soon in the coming modules became an industry standard. And it is also saw up parallelisation and distribution of database systems that, before that database systems were centralized.

That you have one machine, one server or one PC on which you are using it, but in the 80 slowly across enterprise you started using it, you need to use it across the enterprise, you need to use it across a local area network and so on. And in the 90s, as Internet came in this use, or this distribution certainly went all across the web.

In early 2000s, and so on the need for unstructured data management came into being so we had XML and XQuery standards coming in, will talk a little bit about XML and those kinds of stuff slowly and from later 2000s really giant data storage systems, Google Bigtable, Yahoo PNuts, Amazon these kinds of things have started coming.

Converging into the data science possibilities on the, the foundation of data science is not database, it is the foundational part in this is database system, which gives you data warehousing, data mining, all of that, and on top of that you have data science, of which you are trying to gain an expertise. So, think about being a data scientist, among many things, what you will have to be really good at is database management systems.

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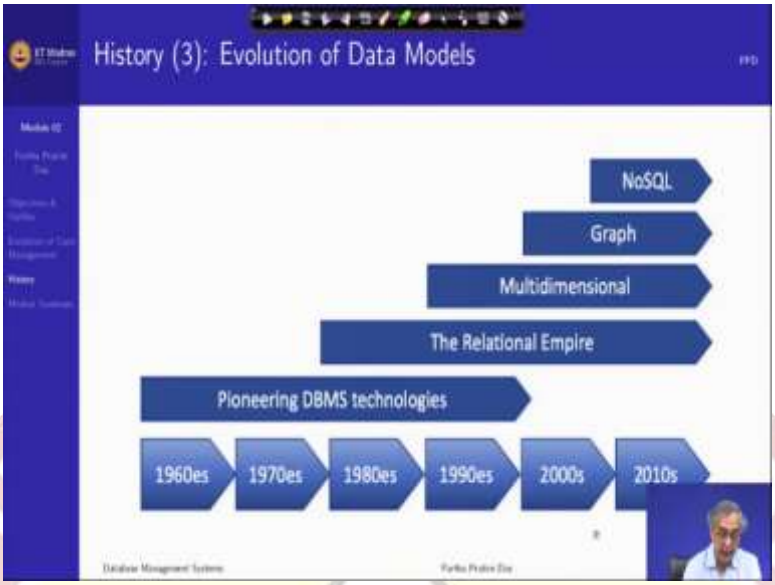

So, there are different evolution models that I have shown here, this is not this is more for, giving a picture about how things have moved from 60's to the 2010. This, you know, horizontal axis show you what has primarily been happening during those decades.

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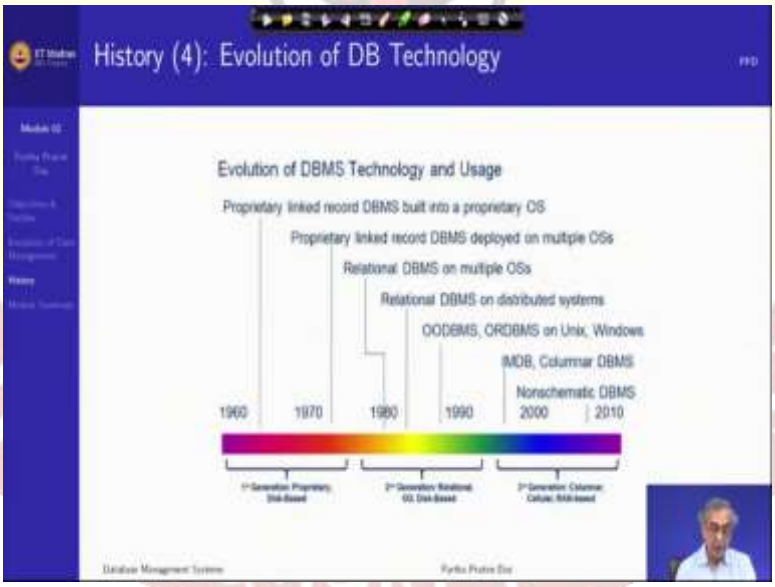

Then how the technology has moved from based on the evolution of the computer technology.

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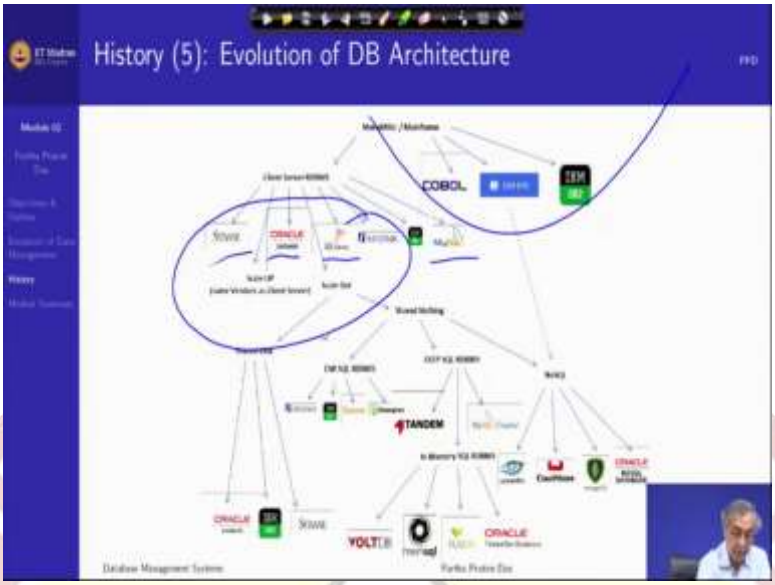

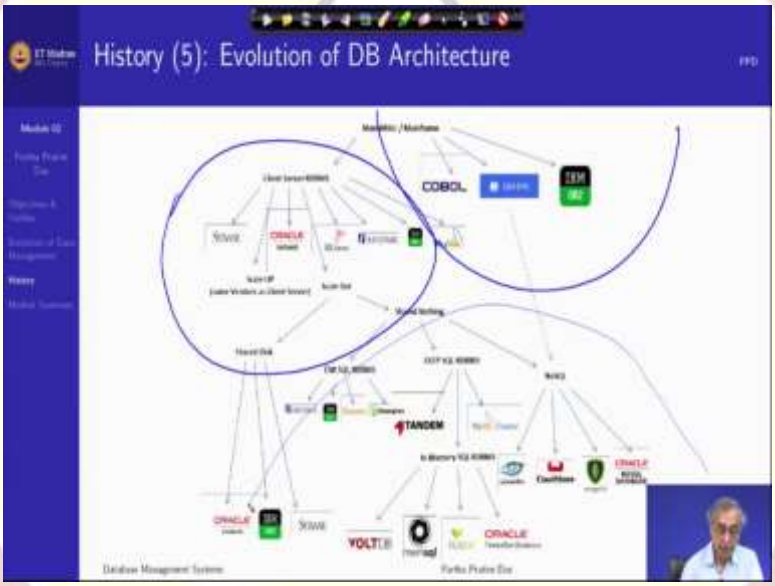

And I will talk about this third part more subsequently, is how the database architecture as move in. And if you if you look at that way, then basically, this is primarily the area of architecture on which we will be focusing, talking about Oracle system, SQL servers, MySQL, Postgre SQL, which is not named here, Sybase, and so on.

And these are what you all see around, this is your past. And this is kind of what has been happening, beyond that I am sorry. Let me choose this. So, this is what we primarily intend to do. This is the past stuff. And this is all different types of what you see here are all different types of systems that are taking things forward, particularly dealing with the big data and so on.

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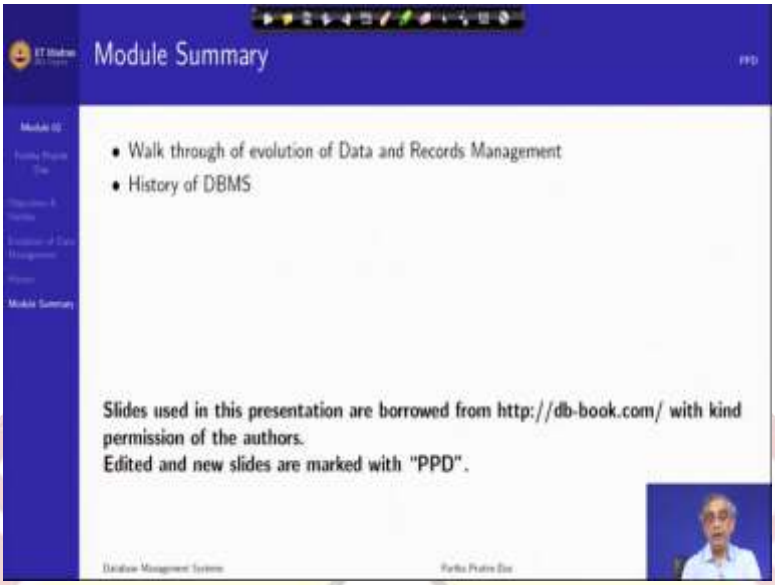

So, that is about the basic overview of how databases been playing a big role in our life, the evolution of the data and records management and the history of the DBMS. Thank you very much for your attention and see you in the next module.